# Lightweight Pothole Detection for Vehicle-Based Road Monitoring

Can a small, fast detector find potholes well enough to report them automatically?

CSE36501 Deep Learning — Project Proposal

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# Motivation

- Today potholes are found by **road-agency inspections** or **citizen reports**.
  - small potholes are rarely reported,
  - potholes on rural / suburban roads are reported late.
- Once autonomous vehicles are common, every car already sees the road $\Rightarrow$ it could detect potholes and report their location automatically.
- But the car's compute is reserved for driving: a monitoring side-task must be efficient, fast, and cheap.

Goal: a pothole detector that is accurate enough *and* light enough to run alongside the driving stack.

## Problem statement

**Task.** Single-class object detection (bounding boxes) of potholes in vehicle-view road images.

**Constraint.** Low compute: we report accuracy *together with* parameters, GFLOPs and inference speed (FPS).

**RQ1.** How small and fast can a pothole detector be while keeping useful accuracy? (accuracy–efficiency trade-off)

**RQ2.** Which efficiency lever costs the least accuracy: a smaller model, a lower input resolution, a lightweight module design, or reduced precision?

**RQ3.** Does an efficient model stay reliable in conditions a car will meet — other countries and cameras, and rain?

# Datasets

| | RDD2022 (Arya et al., 2024) | MWPD (Parvin et al., 2025) |
|---|---|---|
| **Role** | main train / val / test | external robustness test (domain, rain) |
| **Source** | vehicle-mounted cameras & phones, 6 countries | mixed web, news and phone images |
| **Size** | 47,420 images, 4 damage types; we keep **D40 = pothole** (3,610 positive images, 6,458 boxes) | 1,357 images, 640×640 |
| **Labels, license** | Pascal VOC XML; CC BY 4.0 | YOLO txt, 1 class; CC BY 4.0 |
| **Conditions** | clear daylight | normal / rainy; day / twilight / night |

- **Why these:** RDD2022 is the largest public multi-country road-damage set, captured from vehicles — the target viewpoint; MWPD adds a different source and real rain.
- China_Drone excluded (top-down drone view $\neq$ vehicle view).
- MWPD ships without weather labels $\Rightarrow$ every image was tagged normal/rainy (pre-tag + manual review).

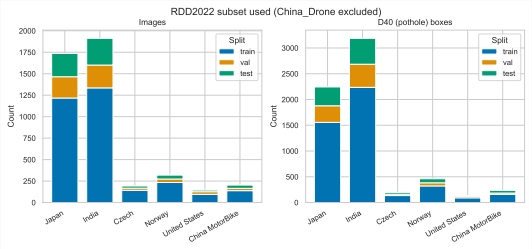
# What the data looks like



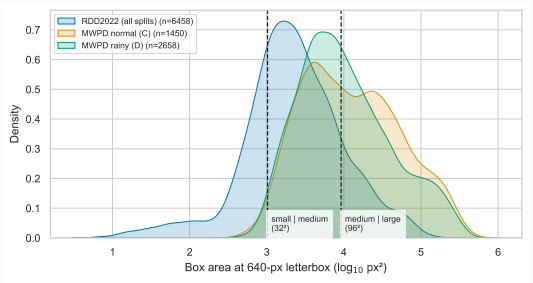

Random test images; green = pothole box (YOLO label).

- **RDD2022:** street scene from a car; potholes are small and far away.
- **MWPD normal:** web close-ups (watermarks); the pothole fills the frame.
- **MWPD rainy:** water-filled, reflective potholes.

# Dataset analysis




Only 10% of annotated RDD images contain a pothole; Japan + India hold 81% of them.



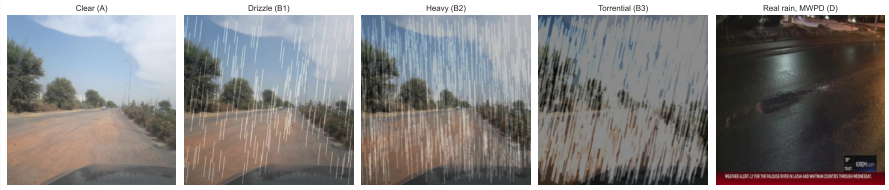

26% of RDD potholes are COCO-small ($< 32^2$ px at 640); MWPD: 1%.

**Why it matters:** small potholes are exactly the ones people do not report — and the first to disappear when we shrink the model or the input resolution.

**Leakage-safe split:** blocks of 50 consecutive frames go whole to train/val/test (70/15/15); negatives capped at 20%.

# Evaluation design

| ID | Composition | Images | Purpose |
|---|---|---|---|
| A | RDD2022 own test split (clear) | 708 | **main accuracy** |
| B1/B2/B3 | *same images* as A + synthetic drizzle/heavy/torrential | 708 each | rain robustness |
| C | MWPD normal weather | 681 | new source / camera |
| D | MWPD rainy | 676 | real rain |




Every model is reported as **accuracy** (mAP50, mAP50-95, recall, small-object AP) **vs. cost** (params, GFLOPs, FPS on GPU and CPU).

B, C, D answer RQ3: does the efficient model still work off the training distribution?

# Plan & timeline

| Step | Content | Status |
|---|---|---|
| 1–2 | Problem, datasets, splits, evaluation sets A–D | done (this talk) |
| 3 | Baselines from tiny to large (YOLOv8n/s, RT-DETR-L as accuracy reference); efficiency levers: input resolution, lightweight own variant, FP16/INT8 | next |
| 4 | Accuracy–cost curves; small-object AP; robustness on B/C/D; 3 seeds | |
| 5 | Analysis, failure cases; report & final video | |

Compute: one RTX 3090 (24 GB) for training; speed also measured on CPU as a proxy for in-vehicle hardware. Everything is scripted and seeded for reproducibility.